CS 173 - Lecture 13

*see discussion problems, lecture explains concepts thru discussion problems

Multiple base cases

ex. claim: every amount of postage at least 12¢ can be made from 4¢ and 5¢ stamps. Prove by induction.

Trying to show $\forall n \geq 12$, $n = 4a + 5b$, $a, b \in \mathbb{N}$ (eg. $12 = 3 \times 4$, $13 = 4 \times 2 + 5$, etc.)

Proof: We will do a proof by induction on the postage amount n. base case 1 Our base case is $n = 12$. We can get 12¢ with three 4¢ stamps, so base case holds. For our inductive hypothesis, suppose we can build a postage amount using 4¢ and 5¢ stamps for $n = 12, 13, 14, \dots, k-1$. For our inductive step, $n = k$, need to show we can build k-cent postage amount using 4 and 5 cent stamps. We know by IH that we can build $(k-4)$ cents using 4/5 cent stamps. If we add a 4¢ stamp, we can build a postage amount of $(k-4) + 4 = k$. $P(12)$ is true for base case, but what about $P(13)$? $P(13-4) = P(9)$, which is undefined. Thus, we can't get $P(13)$ using inductive step. base case 2 Make $P(13)$ another base case, $n = 13$. 13 cents is two 4¢ + one 5¢, so second base case ($n = 13$) is true. What about $P(14)$? Depends on $P(10)$ which is undefined, so make base case 3 base case too, $n = 14$. 14¢ is two 5¢ + one 4¢. base case 4 $P(15)$ needs base case, $n = 15$, 15¢ is three 5¢. Finally, $P(16)$ can use inductive step bc $P(16)$ depends on $P(16-4) = P(12)$, which is defined. □

Takeaway: If IS depends on $k - i$, need at least i base cases

Inequality Induction

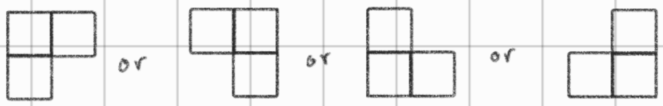
ex. claim: $\sum_{i=2}^{n} \frac{1}{i^2} \leq \frac{3}{4} - \frac{1}{n} \quad \forall \ n \geq 2$

Proof: We will prove $\sum_{i=2}^{n} \frac{1}{i^2} \leq \frac{3}{4} - \frac{1}{n} \ \forall \ n \geq 2$ by induction on n. Our base case is $n = 2$. $\sum_{i=2}^{2} \frac{1}{i^2} = \frac{1}{2^2} = \frac{1}{4}$ on LHS and $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$ on RHS, $\frac{1}{4} \leq \frac{1}{4}$, LHS ≤ RHS, so base case holds. For our inductive hypothesis, assume $\sum_{i=2}^{n} \frac{1}{i^2} \leq \frac{3}{4} - \frac{1}{n}$ holds for $n = 2, 3, 4, \dots, k-1$. Then, our inductive step is where $n = k$ and we will prove $\sum_{i=2}^{k} \frac{1}{i^2} \leq \frac{3}{4} - \frac{1}{k}$. $\sum_{i=2}^{k} \frac{1}{i^2}$ $= \sum_{i=2}^{k-1} \frac{1}{i^2} + \frac{1}{k^2} \leq \frac{3}{4} - \frac{1}{k-1} + \frac{1}{k^2}$ (by IH) $= \frac{3}{4} - \frac{k^2 - (k-1)}{(k-1)k^2} = \frac{3}{4} - \frac{k^2 - k + 1}{k^3 - k^2}$. We are trying to show LHS $\leq \overbrace{\frac{3}{4} - \frac{k^2-k+1}{k^3-k^2}}^{n=k-1} \leq \overbrace{\frac{3}{4} - \frac{1}{k}}^{n=k}$, so $\frac{k^2-k+1}{k^3-k^2} \geq \frac{1}{k}$, $k^3 - k^2 + k \geq k^3 - k^2$, $k \geq 0$ (bc $k > 0$, don't need to flip sign), $k \geq 0$ is true (● = notes, don't include in actual proof). Since $k > 0$, $k^3 - k^2 + k \geq k^3 - k^2$, $\frac{k^2-k+1}{k^3-k^2} \geq \frac{1}{k}$, $\frac{3}{4} - \frac{k^2-k+1}{k^3-k^2} \leq \frac{3}{4} - \frac{1}{k}$. $\sum_{i=0}^{k} \frac{1}{i^2} \leq \frac{3}{4} - \frac{k^2-k+1}{k^3-k^2} \leq \frac{3}{4} - \frac{1}{k}$, so $\sum_{i=0}^{k} \frac{1}{i^2} \leq \frac{3}{4} - \frac{1}{k}$ □

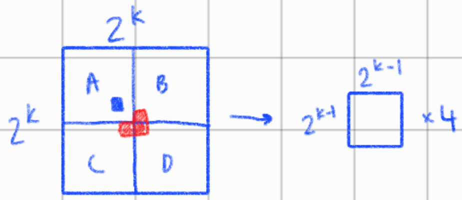
Geometric Induction

Claim: for any positive integer n, a $2^n \times 2^n$ checkerboard with any one square removed can be tiled using right triominoes.

Right trinomino:



Proof: Prove by induction on n. For our base case, $n=1$, so we have a 2×2 grid with one missing square. This is exactly one trinomino. For our inductive hypothesis, suppose we can build $2^n \times 2^n$ grids with one missing square for $n = 1, 2, \dots, k-1$. For our inductive step, we will show we can build a $2^k \times 2^k$ grid with one missing square using triominoes. Break the $2^k \times 2^k$ grid into 4 smaller grids of size $2^{k-1} \times 2^{k-1}$ labeled A, B, C, D. Assume the missing square is in grid A.



A can be built from triominoes by the inductive hypothesis because it is a $2^{k-1} \times 2^{k-1}$ grid with one square missing. Now we take one square out of B, C, D, so now B', C', D' can also be constructed because of the inductive hypothesis. The missing squares from B', C', D' form a trinomino, so we can build the whole $2^k \times 2^k$ grid using triominoes.

Discussion Problems – 11.1c, 11.3, 14.1a

(c) $(\sum_{i=0}^{n} i)^2 = \sum_{i=0}^{n} i^3$ (for all natural numbers)

Proof: We will prove $(\sum_{i=0}^{n} i)^2 = \sum_{i=0}^{n} i^3 \ \forall \mathbb{N}$ by induction on n. For our base case, n = 0, so for the LHS, $(\sum_{i=0}^{0} i)^2 = 0^2 = 0$, and the RHS is $\sum_{i=0}^{0} i^3 = 0^3 = 0$, LHS = RHS so base case holds. For our inductive hypothesis, we assume $(\sum_{i=0}^{n} i)^2 = \sum_{i=0}^{n} i^3$ is true for all $i = 0, 1, 2, \dots, k-1$. Then for our inductive step, we have $n = k$, $(\sum_{i=0}^{k} i)^2 = (0+1+2+\dots+(k-1)+k)^2$

$= (\sum_{i=0}^{k-1} i + k)^2$ (by defn. of summation) $= (\sum_{i=0}^{k-1} i)^2 + 2k(\sum_{i=0}^{k-1} i) + k^2 = \sum_{i=0}^{k-1} i^3 + 2k(\sum_{i=0}^{k-1} i) + k^2$ (by IH)

$= \sum_{i=0}^{k-1} i^3 + 2k\left(\frac{(k-1)k}{2}\right) + k^2$ (bc $\sum_{i=0}^{n} i = \frac{n(n+1)}{2}$) $= \sum_{i=0}^{k-1} i^3 + k^2 + k^2(k-1) = \sum_{i=0}^{k-1} i^3 + k^3 = \sum_{i=0}^{k} i^3$ □

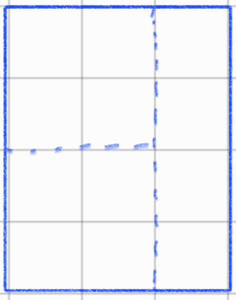
11.3 Geometrical strong induction

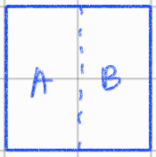
Isaiah is making cupcakes and wants to put a single square of chocolate on top of each. So he is breaking up larger chocolate bars (e.g. 4 squares by 6) into their individual squares. Each break divides the bar in two along a straight line.

For example, we might divide a 4×3 bar into a 4×2 bar and a 4×1 bar. We might then break the larger piece in the other direction, so as to get two 2×2 bars and the 4×1 bar. Nine more breaks are required to finish the process.

Isaiah discovers that an $n \times m$ bar always seems to require $nm - 1$ breaks, apparently regardless of which sequence of directions you choose for the breaks. Use induction to prove that he's right.

Hint: Your induction variable should be a single integer that measures the size of the bar. Be sure to explain what this variable is (in terms of the chocolate bars) at the start of your proof.





Proof: We will prove by induction on x where x is the number of squares and is equal to $n*m$, the height and width in squares, respectively. For our base case, we'll take $x = 1$, where the bar is just one square. $nm - 1 = 1 \times 1 - 1 = 0$, which logically makes sense because there are zero ways to break only one square. For our inductive hypothesis, we suppose $nm - 1 = x - 1 = \#$ of breaks to break a bar with x squares is true for $x = 1, 2, \dots, k-1$. For our inductive step, we're trying to prove that $nm - 1 = x - 1 = \#$ breaks for $x = k$, so $k - 1 = \#$ breaks. We will break our bar with k squares into two pieces – A and B. Let A have w squares and B have z squares, where $w + z = k$. By the inductive hypothesis, w requires $w - 1$ and B requires $z - 1$ breaks. If we account for the break between A and B, we get $\#$ breaks $= (w-1) + (z-1) + 1 = (w+z) - 1 = k - 1$ □

14.1 Induction with Inequalities

(a) Prove that $n^2 > 7n + 1$ for all integers $n \geq 8$

$7k - 1 < k^2$ $\quad$ $6k + k - 1 < k^2$ $\quad$ $9k - 7 > 7k + 1$

$(k-1)^2 > 7(k-1) + 1$ $\quad$ $(k-1)^2 > 7k - 6$ $\quad$ $2k > 8 \quad k > 4$

$k^2 - 2k + 1 > 7k - 6$

$k^2 > 9k - 7$

Proof: We will prove $n^2 > 7n + 1 \ \forall (n \in \mathbb{Z} \wedge n \geq 8)$ by induction on n. For our base case, $n = 8$, so $8^2 > 7(8) + 1 \rightarrow 64 > 56 + 1 \rightarrow 64 > 57$, which is true so the base case holds. For our inductive hypothesis, we suppose $n^2 > 7n + 1$ is true for $n = 8, 9, 10, \dots, k-1$. For our inductive step, we're trying to prove $n^2 > 7n + 1$ is true for $n = k$ where $k \geq 8$, so $k^2 > 7k + 1$. From our inductive hypothesis, we know that $(k-1)^2 > 7(k-1)-1$, which rearranges to $k^2 > 9k-7$.

$k>4$, so $2k>8$, $9k-7>7k-1$, $k^2>9k-7>7k-1$, so $k^2>7k-1$ $\square$